

WT Experiment:5

Soham Satpute/51

Prajyot Shinde/55

Laksh Sodhai/57

Aim : To design an ultrasonic radar system using Arduino for real-time object detection and distance measurement.

Theory:

A radar system using an ultrasonic sensor and a servo motor operates on the principle of echolocation, similar to how bats and sonar systems detect objects. The **HC-SR04 ultrasonic sensor** emits ultrasonic waves, which reflect off objects and return to the sensor. By measuring the time taken for the echo to return, the system calculates the distance to the object.

A **servo motor** rotates the ultrasonic sensor within a defined range (e.g., 0° to 180°), allowing the system to scan the surroundings. The Arduino Uno acts as the central controller, processing sensor data and displaying the detected objects on a screen (e.g., a serial monitor or a graphical interface).

This project demonstrates fundamental radar-like object detection, useful in applications such as security surveillance, obstacle avoidance, and automation.

Working:

1. **Sensor Initialization:** The **HC-SR04 ultrasonic sensor** is mounted on a **servo motor** and connected to the **Arduino Uno**. The system is powered via USB or a 9V battery.
2. **Servo Rotation:** The **servo motor rotates** the ultrasonic sensor step by step (e.g., from 0° to 180° and back). This movement simulates a radar scanning motion.
3. **Distance Measurement:** At each angle, the **ultrasonic sensor emits a pulse** of high-frequency sound waves. When these waves hit an obstacle, they bounce back and are detected by the sensor.
4. **Data Processing:** The **Arduino calculates** the distance to the object based on the time difference between wave emission and echo reception using the formula:

$$\text{Distance} = \frac{\text{Time} \times \text{Speed of Sound}}{2}$$

where the speed of sound in air is approximately **343 m/s**.

5. **Visualization:** The detected distance and angle data can be displayed on:
 - **Serial Monitor** (text output)
 - **Processing or Python GUI** (graphical radar visualization)
6. **Loop Execution:** The servo continuously sweeps back and forth, updating the detection data in real time.

Steps for Simulating the Ultrasonic Radar System:

Step 1: Open the Simulation Platform

- Use **Tinkercad**, **Proteus**, or any Arduino simulation software.
- Create a **new project** in the simulation environment.

Step 2: Add Components

- Drag and place the following components onto the simulation workspace:
 1. **Arduino Uno** (Microcontroller)
 2. **HC-SR04 Ultrasonic Sensor** (For distance measurement)
 3. **Servo Motor** (For sensor rotation)
 4. **Breadboard** (For easy wiring)
 5. **Jumper Wires** (For electrical connections)
 6. **Power Source** (USB or 9V Battery)

Step 3: Connect the Components

- **Servo Motor:**
 - Connect **VCC** to **5V** on Arduino
 - Connect **GND** to **GND**
 - Connect **Signal (PWM)** to **Digital Pin 9**
- **Ultrasonic Sensor (HC-SR04):**
 - **VCC** → **5V** on Arduino
 - **GND** → **GND**
 - **Trig** → **Digital Pin 10**
 - **Echo** → **Digital Pin 11**

Step 4: Write the Arduino Code

- Open **Arduino IDE** and upload the following code:

```
include <Servo.h>
```

```
// Defines Trig and Echo pins of the Ultrasonic Sensor
const int trigPin = 5;
const int echoPin = 6;
long duration;
int distance;
```

```
Servo myServo; // Creates a servo object
```

```
void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
  myServo.attach(7); // Servo motor connected to pin 7
}
```

```

    }

void loop() {
  // Rotates the servo motor from 15 to 165
  degrees (slowly)
  for (int i = 15; i <= 165; i += 2) { // Moves in
    steps of 2 instead of 1
    myServo.write(i);
    delay(50); // Increased delay for slower
    movement
    distance = calculateDistance();

    Serial.print("Angle: ");
    Serial.print(i);
    Serial.print("°, Distance: ");
    Serial.print(distance);
    Serial.println(" cm");
  }

  // Rotates back from 165 to 15 degrees
  (slowly)
  for (int i = 165; i > 15; i -= 2) { // Moves in
    steps of 2
    myServo.write(i);
    delay(50); // Increased delay for smoother
    movement
    distance = calculateDistance();

    Serial.print("Angle: ");
    Serial.print(i);
    Serial.print("°, Distance: ");
    Serial.print(distance);
    Serial.println(" cm");
  }
}

// Function to calculate distance using
Ultrasonic Sensor
int calculateDistance() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);

```

```

    distance = duration * 0.034 / 2;
    return distance;
  }

```

Processing Code for Radar Visualization:

```

import processing.serial.*; // imports library
for serial communication
import java.awt.event.KeyEvent; // imports
library for reading the data from the serial port
import java.io.IOException;
Serial myPort; // defines Object Serial
// defubes variables
String angle="";
String distance="";
String data="";
String noObject;
float pixsDistance;
int iAngle, iDistance;
int index1=0;
int index2=0;
PFont orcFont;
void setup() {

  size (1200, 700); // ***CHANGE THIS TO
  YOUR SCREEN RESOLUTION***
  smooth();
  myPort = new Serial(this,"COM5", 9600); //
  starts the serial communication
  myPort.bufferUntil('.'); // reads the data from
  the serial port up to the character '.'. So actually
  it reads this: angle,distance.
}

void draw() {

  fill(98,245,31);
  // simulating motion blur and slow fade of the
  moving line
  noStroke();
  fill(0,4);

```

```

rect(0, 0, width, height-height*0.065);

fill(98,245,31); // green color
// calls the functions for drawing the radar
drawRadar();
drawLine();
drawObject();
drawText();
}
void serialEvent (Serial myPort) { // starts
reading data from the Serial Port
    // reads the data from the Serial Port up to the
character '.' and puts it into the String variable
"data".
    data = myPort.readStringUntil('.');
    data = data.substring(0,data.length()-1);

    index1 = data.indexOf(","); // find the
character ',' and puts it into the variable
"index1"
    angle= data.substring(0, index1); // read the
data from position "0" to position of the
variable index1 or thats the value of the angle
the Arduino Board sent into the Serial Port
    distance= data.substring(index1+1,
data.length()); // read the data from position
"index1" to the end of the data pr thats the
value of the distance

    // converts the String variables into Integer
    iAngle = int(angle);
    iDistance = int(distance);
}
void drawRadar() {
    pushMatrix();
    translate(width/2,height-height*0.074); //
moves the starting coordinats to new location
    noFill();
    strokeWeight(2);
    stroke(98,245,31);
    // draws the arc lines
    arc(0,0,(width-width*0.0625),(width-
width*0.0625),PI,TWO_PI);
    arc(0,0,(width-width*0.27),(width-
width*0.27),PI,TWO_PI);
    arc(0,0,(width-width*0.479),(width-
width*0.479),PI,TWO_PI);

```

```

    arc(0,0,(width-width*0.687),(width-
width*0.687),PI,TWO_PI);
    // draws the angle lines
    line(-width/2,0,width/2,0);
    line(0,0,(-width/2)*cos(radians(30)),(-
width/2)*sin(radians(30)));
    line(0,0,(-width/2)*cos(radians(60)),(-
width/2)*sin(radians(60)));
    line(0,0,(-width/2)*cos(radians(90)),(-
width/2)*sin(radians(90)));
    line(0,0,(-width/2)*cos(radians(120)),(-
width/2)*sin(radians(120)));
    line(0,0,(-width/2)*cos(radians(150)),(-
width/2)*sin(radians(150)));
    line((-width/2)*cos(radians(30)),0,width/2,0);
    popMatrix();
}
void drawObject() {
    pushMatrix();
    translate(width/2,height-height*0.074); //
moves the starting coordinats to new location
    strokeWeight(9);
    stroke(255,10,10); // red color
    pixsDistance = iDistance*((height-
height*0.1666)*0.025); // covers the distance
from the sensor from cm to pixels
    // limiting the range to 40 cms
    if(iDistance<40){
        // draws the object according to the angle
and the distance
        line(pixsDistance*cos(radians(iAngle)),-
pixsDistance*sin(radians(iAngle)),(width-
width*0.505)*cos(radians(iAngle)),-(width-
width*0.505)*sin(radians(iAngle)));
    }
    popMatrix();
}
void drawLine() {
    pushMatrix();
    strokeWeight(9);
    stroke(30,250,60);
    translate(width/2,height-height*0.074); //
moves the starting coordinats to new location
    line(0,0,(height-
height*0.12)*cos(radians(iAngle)),-(height-
height*0.12)*sin(radians(iAngle))); // draws
the line according to the angle

```

```

    popMatrix();
}
void drawText() { // draws the texts on the
screen

    pushMatrix();
    if(iDistance>40) {
        noObject = "Out of Range";
    }
    else {
        noObject = "In Range";
    }
    fill(0,0,0);
    noStroke();
    rect(0, height-height*0.0648, width, height);
    fill(98,245,31);
    textSize(25);

    text("10cm",width-width*0.3854,height-
height*0.0833);
    text("20cm",width-width*0.281,height-
height*0.0833);
    text("30cm",width-width*0.177,height-
height*0.0833);
    text("40cm",width-width*0.0729,height-
height*0.0833);
    textSize(40);
    text("ELECTRICAL MANDIR ", width-
width*0.875, height-height*0.0277);
    text("Angle: " + iAngle + " ", width-
width*0.48, height-height*0.0277);
    text("Distance: ", width-width*0.26, height-
height*0.0277);
    if(iDistance<40) {
        text("      " + iDistance +" cm", width-
width*0.225, height-height*0.0277);
    }

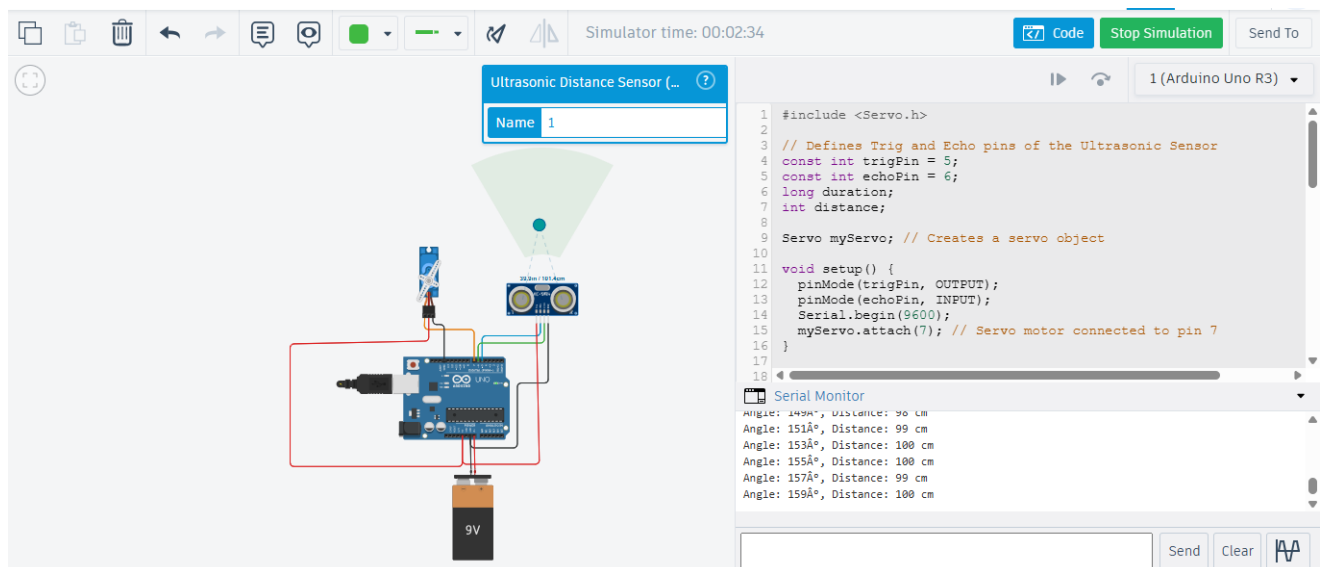
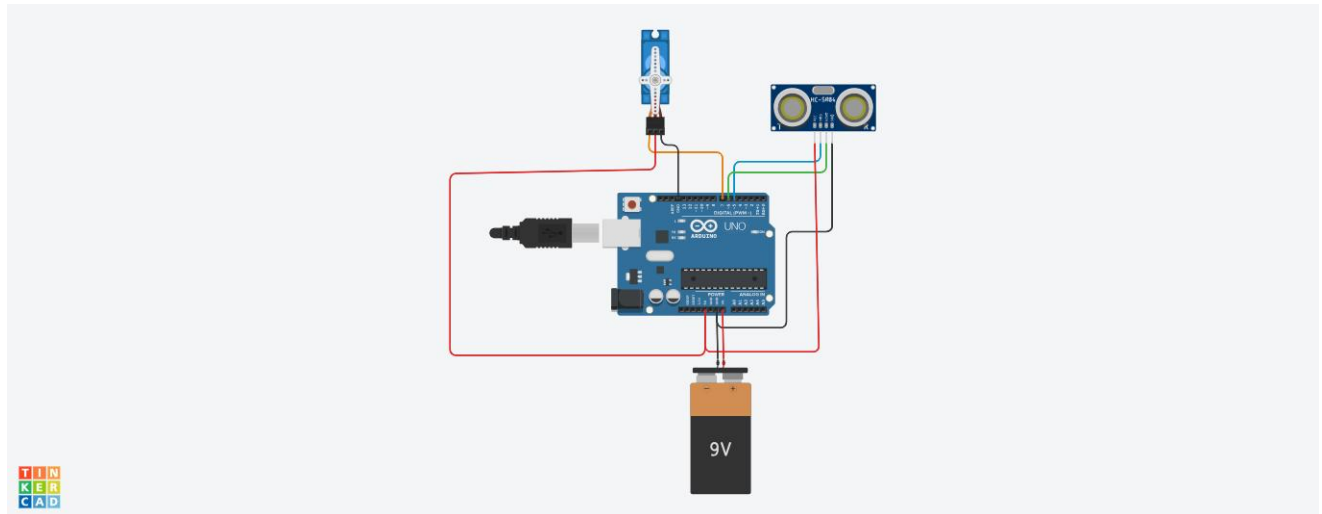
```

```

    textSize(25);
    fill(98,245,60);
    translate((width-
width*0.4994)+width/2*cos(radians(30)),(heig
ht-height*0.0907)-width/2*sin(radians(30)));
    rotate(-radians(-60));
    text("30 ",0,0);
    resetMatrix();
    translate((width-
width*0.503)+width/2*cos(radians(60)),(heigh
t-height*0.0888)-width/2*sin(radians(60)));
    rotate(-radians(-30));
    text("60 ",0,0);
    resetMatrix();
    translate((width-
width*0.507)+width/2*cos(radians(90)),(heigh
t-height*0.0833)-width/2*sin(radians(90)));
    rotate(radians(0));
    text("90 ",0,0);
    resetMatrix();
    translate(width-
width*0.513+width/2*cos(radians(120)),(heig
ht-height*0.07129)-
width/2*sin(radians(120)));
    rotate(radians(-30));
    text("120 ",0,0);
    resetMatrix();
    translate((width-
width*0.5104)+width/2*cos(radians(150)),(hei
ght-height*0.0574)-
width/2*sin(radians(150)));
    rotate(radians(-60));
    text("150 ",0,0);
    popMatrix();
}

```

Output :



Conclusion:

The **Arduino-based Radar System** successfully detects and visualizes objects using an **ultrasonic sensor (HC-SR04)** mounted on a **servo motor**. The system continuously scans a **180-degree area**, measuring distances and displaying the data in a **Processing IDE-based radar interface**.

Through **serial communication**, the Arduino sends real-time **angle and distance data** to the Processing software, which generates a graphical representation of detected objects. The project effectively demonstrates the **integration of embedded systems and real-time data visualization**, making it useful for applications such as **proximity sensing, obstacle detection, and security systems**.